Dynamic Importance Weights in Reinforcement Learning with ChatGPT for Drone Swarms in Search and Rescue

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ABSTRACT

In the evolving field of disaster search and rescue, drone swarms have emerged as a crucial technology for efficient and rapid response. However, the inability of existing systems to dynamically adapt to diverse tasks poses a major limitation. Specifically, current drone technology lacks real-time adaptability to changing task priorities, which is essential in the unpredictable and varied environments of disaster scenarios. This adaptability is crucial for navigating complex terrain, perceiving through natural barriers, and coordinating communication efficiently amongst drones. This paper aims to develop a drone swarm system capable of autonomously adjusting the priorities of these tasks in response to different disaster scenarios. This paper will propose the development of a drone swarm system designed to autonomously adjust task priorities in response to different disaster scenarios, utilizing a novel reinforcement learning approach. Initially, all drones are assigned uniform weights, which are subsequently varied among certain drones under different disaster conditions. By analyzing the system's overall performance and the drones' collaborative behavior, we aim to identify the optimal mix of drones with varied task priorities. Our research is expected to significantly enhance the adaptability and effectiveness of drone swarms in disaster scenarios, leading to more responsive and agile search and rescue operations.

KEYWORDS: Multi-objective optimization; swarm; search and rescue

INTRODUCTION

Unmanned Aerial Vehicles (UAVs) or drones have gained significant popularity due to their efficiency, versatility, and flexibility in various mission spaces. One of these growing applications is to support search and rescue (SAR), given drones' capacity to navigate challenging disaster-stricken environments (Mishra et al. 2020; Restas 2015). Especially the advances in autonomous drones equipped with advanced control systems and sensors, such as LiDAR and thermal imaging sensors, have revolutionized SAR efforts(You et al. 2023). Autonomous drones can efficiently collect data while in flight, supporting tasks like rapid damage assessment, survivor detection, and situational awareness in disaster-affected areas (Rohman et al. 2019) (Esposito et al. 2021) (Entrop and Vasenev 2017). Moreover, autonomous drones are capable of planning flight paths without human intervention, allowing them to adapt to dynamic environmental conditions that are often encountered during disaster response efforts. In the chaotic aftermath of disasters, the ability to reach remote or hazardous locations quickly is crucial for effective response. Autonomous drones can navigate through debris, unstable terrain, and challenging weather conditions, enabling them to provide critical assistance where it is needed most (Ragi and Chong 2013). This autonomy also plays a vital role in relieving human responders from repetitive and potentially hazardous tasks, ultimately improving overall efficiency and safety during disaster search and rescue operations (Shavarani et al. 2018).

Drone swarms, operating in coordination, significantly enhance search and rescue operations, leveraging their collective capabilities beyond individual drone functions(Arnold et al. 2018). However, a key challenge in their effective deployment in disaster scenarios is their lack of dynamic adaptability, a limitation of current autonomous systems (Arnold et al. 2022). In the challenging contexts of disaster response, the flexibility of drone swarms to adjust and select task priorities dynamically is vital, yet current methods fall short of providing this crucial feature. This ability to modify priorities in real time is essential for the effective operation of drones in diverse and unpredictable scenarios. For instance, a drone swarm faced with a choice between searching for survivors in a collapsing structure or delivering emergency aid must make this decision swiftly, adapting to rapidly evolving conditions – a capability that current RL-based systems often lack. Current RL-based systems in drone technology often rely on algorithms with manually pre-set priorities. While this approach may have some applications, it falls short in diverse and unpredictable disaster scenarios. This rigidity hinders their capacity for autonomous, real-time adaptation to the ever-changing conditions often encountered at disaster sites. In disaster scenarios, where situations can shift suddenly, the ability of a swarm to quickly reorganize and adapt to new priorities becomes crucial (Chen and Rho 2020). In the current state, drone swarm technology often focuses on a singular high-priority task, like navigating complex terrains or detecting survivors in debris (Haddad et al. 2023). A robust decision-making framework, capable of intelligent behavior based on real-time data, is needed to support this adaptability. Addressing these issues, our research introduces a novel system employing GPT for real-time, adaptive priority decision-making in drone swarms. This system is designed to dynamically adjust task priorities based on the specific and evolving demands of the disaster environment, aiming to enhance the effectiveness and responsiveness of operations in diverse and rapidly changing disaster situations.

In this study, we introduce an innovative method to enhance the training of autonomous drone swarm systems. This is achieved by incorporating cutting-edge reinforcement learning techniques. We aim to develop a drone swarm system that can autonomously and intelligently adapt to diverse conditions encountered in rescue operations. This system is expected to heighten the drone swarm's overall efficiency and offer deeper insights into the most effective strategies for deploying drones in intricate and ever-changing rescue scenarios. During the training, we use the ChatGPT as the meta-agent of the reinforcement learning process, whose mission is to gather real-time environmental information and then decide the priorities of all the tasks. After the training, the drone swarm system can show different behaviors during the different phases of a complex task than the swarm system trained by a traditional reinforcement learning system.

RELATED WORKS

The application of drone technology in disaster search and rescue operations introduces a new dimension of challenges, particularly in multi-objective task planning. This includes the need for drones to autonomously adjust their priorities based on the rapidly changing demands of disaster scenarios. Recent studies suggest the need for more advanced, adaptable systems. Kyriakakis et al. developed the Moving Peak Drone Search Problem (MPDSP) for dynamic optimization in

UAV search and rescue, using a multi-swarm framework to test seven algorithms (Kyriakakis et al. 2021). They used peak functions to generate a map of position values in terms of difficulty in finding the target and also the value of finding the target to determine the sequence of the targets.

Cardona et al. developed a distributed victim-detection algorithm for drones using CNNs in search and rescue missions, enabling drone swarms for victim validation and employing rendezvous consensus for information gathering (Cardona et al. 2021). They allowed the main swarm to form a new sub-swarm around the victim whenever they found a new victim. Wei et al. introduced a novel multi-objective particle swarm optimization method for cooperative robot task allocation in SAR, aimed at minimizing total team cost and balancing workload (Wei et al. 2020). Hayat et al. developed two adaptive strategies, SIC with QoS (SICQ) and SIC following QoS (SIC+), for multi-UAV path planning in search and rescue missions (Hayat et al. 2020). These strategies integrated communication as a mission goal, not just a constraint, to enable dynamic task allocation. They focused on optimizing tasks like searching, informing, and monitoring, ensuring link quality without compromising area coverage or mission time. Du proposed an enhanced path-planning method for UAVs in SAR missions by integrating the A* algorithm with a task allocation algorithm (Du 2023). This approach aimed to reduce calculation time and memory usage, effectively handling cooperative tasks. The study focused on optimizing the A* algorithm to create shorter, collision-free drone swarm paths. These traditional methods, often relying on fixed algorithms or operator input, fall short in these high-stakes, unpredictable settings. These gaps in current technology underscore the importance of our research in developing innovative approaches for dynamic priority setting in drone swarms, which is crucial for effective disaster management.

Literature has also explored various advanced reinforcement learning algorithms for multi-agents in SAR conditions. Horyna et al. developed a distributed autonomous flocking behavior for UAVs, utilizing decentralized swarm navigation and real-time RGB camera detections for search and rescue applications (Horyna et al. 2023). Sun et al. proposed DELTAS, a decentralized reinforcement learning strategy for UAV teams, aimed at enhancing multi-agent cooperative search in partially known environments with uncertain target locations. Their approach, addressing the limited capabilities of both searchers and targets, featured a network design that stabilized the learning process and reduced bias in value estimation, with its effectiveness confirmed through simulated experiments (Sun et al. 2021). However, existing methods still use static weights to train the RL model. These methods may face challenges in dynamic task allocation and adapting to uncertain conditions, highlighting the need for further research in this area.

In this study, we integrated meta-learning with reinforcement learning to train drone swarms for multi-objective tasks in search and rescue scenarios. We focused on situations where target locations, like survivors, are initially unknown, requiring the drones to search and reach these locations quickly. Given the challenge of pre-defining objective priorities, which depend on real-time environmental conditions, we proposed using an LLM model, like ChatGPT, to oversee the reinforcement learning process. This approach aims to optimize the drones' path planning and decision-making in these dynamic and unpredictable environments.

METHODOLOGY

Overview. The present research introduces a novel approach combining meta-learning and reinforcement learning for drone swarms in search and rescue missions. Our method is designed for situations where the locations of targets, such as survivors, are unknown, necessitating rapid search and precise localization by the drones. We propose the integration of an LLM model like ChatGPT to oversee and optimize the reinforcement learning process, enhancing the drones' path planning and decision-making abilities in real time. This approach allows for adaptive, efficient navigation in complex, unpredictable environments, addressing the challenge of dynamically prioritizing objectives based on ever-changing conditions.

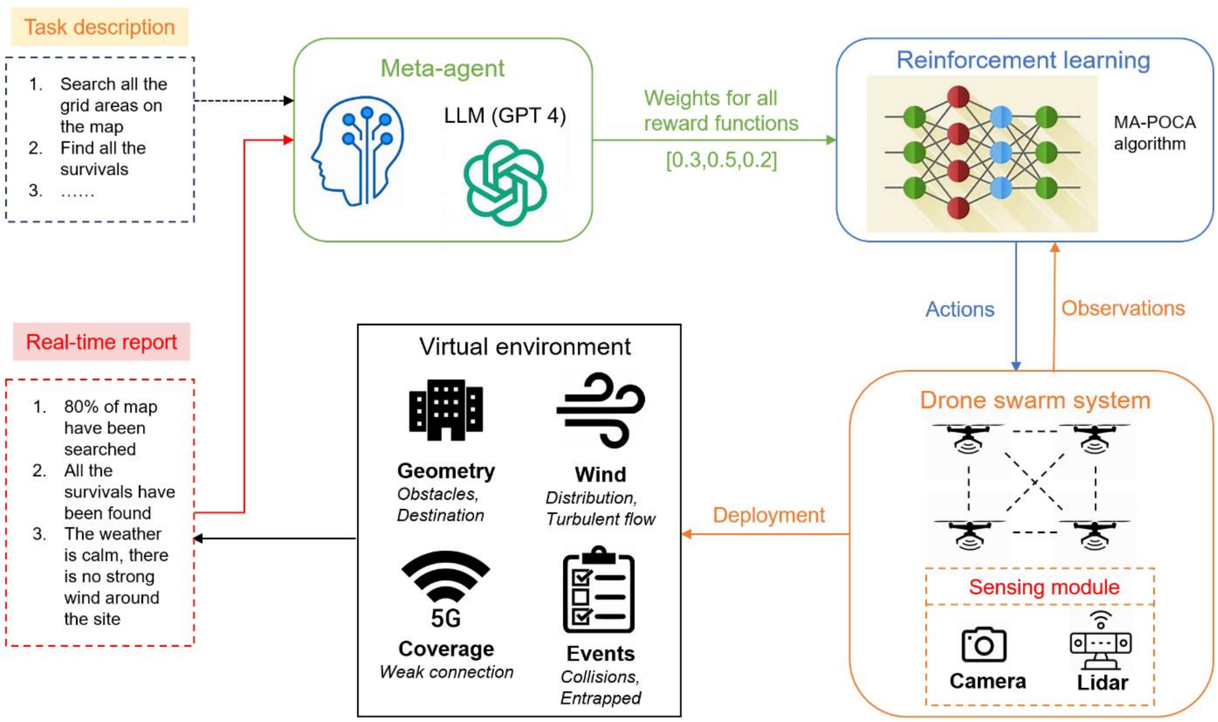

Figure 1. The framework of the proposed method

The proposed method is presented in Figure 1. Before the training, a task description will be sent to the meta-agent, which is the ChatGPT in our study. Its primary function is to enable rapid adaptation and efficient learning for new or changeable tasks, using the knowledge gained from past experiences. For the reinforcement learning part, a multivariate reward function is utilized to train the drone swarm system to finish the search and retrieve tasks while avoiding all the obstacles. After every certain step, the drone agent will send a real-time report to the LLM about the completeness of all the tasks and if there is some emergency things happen. Then, based on this information, the meta-agent will change the priorities of all the tasks, which are the weights of the rewards function in the reinforcement learning section.

In general, its decision network for the drone swarm system can be defined as follows:

$$D_t = F(o_t, v_t, w_t, p_t, W_t, D_{t-n})$$

Where D_t is the speed decision for this step, o_t represents the obstacle data (relative position) from the lidar sensor, v_t represents the current speed of the drone, w_t is the wind data based on the drifting of the drone, p_t refers to the local positions of all drones in swarm system, which are used to control different drone fly in the different designed scope to accelerate the speed of the search, W_t represents the weights for different tasks at this step, and D_{t-n} are the previous decisions, which are memorized by the LSTM structure.

In this experiment, the reward function is formulated and represented as follows:

$$R = r_{map} + r_{target} + r_{collision} + r_{stuck} + r_{time} + r_{timeout}$$

Where R is a combination of different reward terms that the drone can receive, r_{map} is a reward based on how many gird areas the drone has searched on the map, r_{target} is a binary reward indicating whether the drone successfully finds the target and safely reaches the target or not, $r_{collision}$ is a penalty incurred when the drone collides with any obstacles, r_{stuck} is a penalty imposed when the drone keep searching the area that has been explored during the training, r_{time} is a penalty term based on the total time taken for search and rescue tasks, $r_{timeout}$ is a binary term representing a penalty when the drone runs out of time.

Meta-learning. Meta-learning, a progressive field in machine learning, is central to our research on dynamically modifying the weights of the reward function in reinforcement learning algorithms. This concept, known as "learning to learn," empowers algorithms to adapt their learning processes to new and varied tasks with minimal data. In our work, we harness the principles of meta-learning to enable our model to intelligently adjust the weights of its reward function, a process essential for tasks where learning environments and objectives are not static. In traditional reinforcement learning, a fixed reward function often limits a model's ability to adapt to changing scenarios. However, by integrating meta-learning, our model not only learns a specific task but also develops an understanding of how to modify its learning strategy - in this case, the reward function - according to the dynamics of the environment. This adaptability is crucial in environments where conditions and goals continuously evolve, requiring the model to reassess and re-prioritize its learning objectives. Our approach extends the meta-learning framework to specifically focus on the dynamic adjustment of reward function weights. This allows our model to perform effectively across a variety of tasks and conditions, making it particularly suited for complex, real-world applications where flexibility and rapid adaptation are key. The incorporation of meta-learning into reinforcement learning for dynamic reward function adjustment represents a significant advancement in the field, offering a more nuanced and responsive approach to machine learning challenges.

Multi-agent reinforcement learning. We used the MA-POCA algorithm to train the drone swarm system(Cohen et al. 2021). It is a specialized approach for multi-agent reinforcement learning that is particularly adept at handling environments where the number of agents can change, such as agents joining or leaving a task, which is very suitable for the drone swarm system. This is a significant enhancement over previous models, which often require a fixed number of agents. MA-POCA introduces a novel approach to the credit assignment problem. Typically, in multi-agent reinforcement learning, it's challenging to determine how much an individual agent's action has contributed to the overall outcome, especially when agents might be terminated early and not directly observe the outcomes of their actions. MA-POCA addresses this by propagating value to the agents even after they have been terminated, allowing for an assessment of the actions' long-term impact on the team's success. When it comes to shared experiences and learning, MA-POCA pools observations from multiple agents to learn in a centralized manner while managing individual and collective rewards. This pooling is crucial for coordinated actions and learning from the shared experiences of the group, which helps improve the overall efficiency of the learning process. Overall, MA-POCA represents a significant advancement in multi-agent reinforcement learning, offering a robust framework for environments with dynamic agent populations and complex interaction patterns.

EXPERIMENT

We expanded our previous work, which was about single autonomous drone navigation by using the PPO algorithm, to build up this virtual simulation environment(Wu et al. 2024). We applied the same configurations for MA-POCA as those for PPO because there are no additional hyperparameters between MA-POCA and PPO. In our study, we created simulations of various urban districts, each measuring between 1500 and 3000 feet. These districts were segmented into 25 grid areas, with each grid area being 300 x 300 feet in size. Prior to the task, a varying number of targets are randomly placed on the map, unknown to the drones. During training, the drone swarm must learn to collaboratively search the entire map efficiently and navigate safely to specific targets. The simulation includes 50 buildings of various sizes in each epoch as obstacles. To add realism, we enabled 3D navigation with a height cap of 160 feet, preventing drones from simply flying above the buildings.

To accurately simulate real-world drone behavior, our simulation uses parameters consistent with typical consumer-grade drones. These drones usually weigh between 0.55 and 4.4 pounds and can achieve speeds of 20 to 60 mph, depending on the model. They are often equipped with cameras of varying quality and have maximum altitude limits based on specifications and local regulations. In our simulation, the drones are modeled with a weight of 2 pounds, a top speed of 30 mph, and a maximum altitude of 160 feet. Additionally, they are equipped with a Lidar sensor to detect the obstacles around them and a RGB camera to identify the survivors.

In our drone swarm system, a single agent will control all drones to enhance their cooperative capabilities. Each drone is equipped with GPS and shares a global map with the swarm. This map includes information on previously searched grid areas and the locations of targets, facilitating efficient coordination and task execution.

RESULTS

We conducted a series of evaluations to assess the effectiveness of our method compared to the traditional reinforcement learning methodology in directing the drone swarm system to thoroughly search the district and safely reach all targets. For visual clarity in our results, each target is represented as an orange sphere on the map, while the grid areas are depicted as blue squares. Additionally, to distinctly illustrate each drone’s trajectory, we assigned a unique color to represent the route of every drone.

Figure 2 illustrates the training process of the drone swarm system for simultaneous search and rescue tasks by using a traditional reinforcement learning algorithm. Initially, as shown in Figures 2(a) and (b), the drones exhibited random exploration patterns and frequently collided with the district’s boundaries, reflecting their lack of task understanding and initial unfamiliarity with the data from the Lidar sensors. As depicted in Figure 2(c), the drones had learned their objectives and started to navigate toward nearby grids, motivated by the assigned reward system. However, as depicted in Figure 2(d), the drone swarm system encountered difficulties when searching for the final grid of the map. Multi-objective tasks in reinforcement learning typically demand substantial data inputs. This is particularly true for data from Lidar sensors, which provide dense point cloud information. Consequently, this requirement for extensive data can significantly slow down the convergence rate of the reinforcement learning process, potentially leading to outcomes like those shown in Figure 2(d).

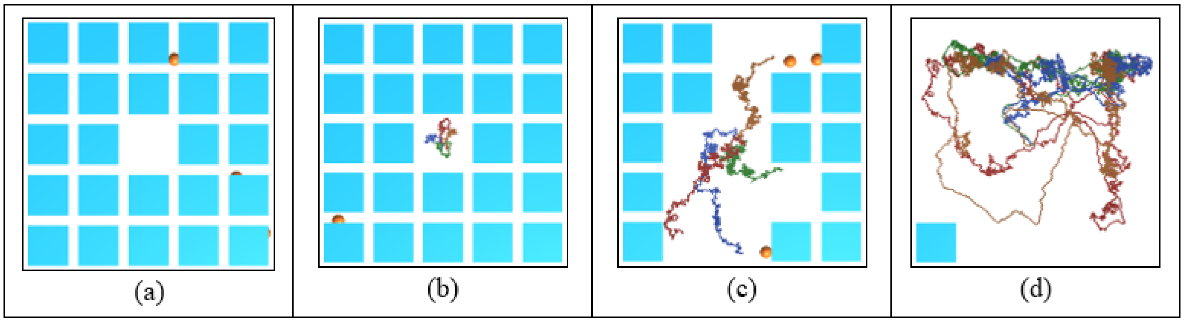

Figure 2. Examples of training results in a map with Traditional RL

Figure 3 presents the outcomes of the training process facilitated by ChatGPT. Figure 3(a) shows that the agent learned to deploy drones in different directions to expedite the search process. The agent directed each of the four drones to distinct areas, enabling simultaneous comprehensive map coverage. Figure 3(b) illustrates the results when dynamic weights are applied to tasks. In this scenario, with all targets already located, only one grid remained to be searched. ChatGPT adjusted the search weights accordingly, prompting the agent to dispatch a drone to swiftly survey the last grid, as indicated in the figure. Figure 3(c) exhibits the coordinated behavior of the drone swarm system, highlighting their efficiency in completing all tasks in a remarkably short duration. Table 1 summarizes all the weights assigned by ChatGPT throughout the training process; the first number represents the weight for arriving at the targets, which is the rescue and the second weight represents the weight for searching the whole map. Finally, Figure 3(d) demonstrates that all the drones rapidly adapt to environments with obstacles and collaborate effectively to execute the search and rescue tasks, with a success rate of 95% after 100,000 epochs.

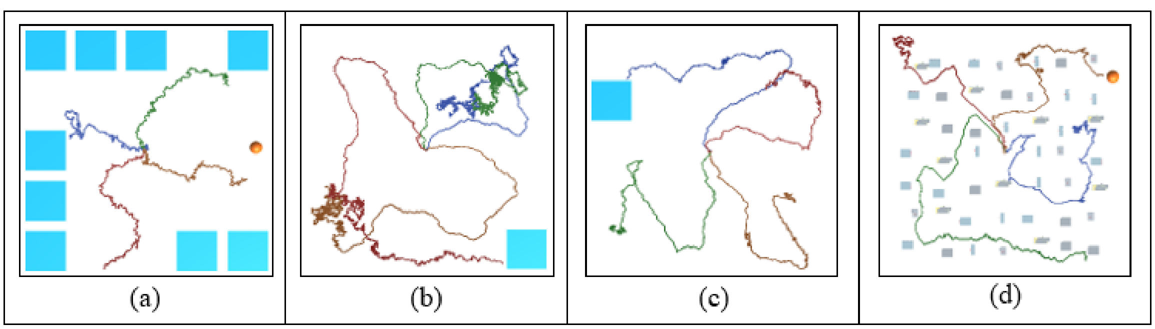

Figure 3. Examples of training results in a map with ChatGPT interaction

Figure 4 depicts the progression of mean rewards over training steps for the drone swarm system with and without the assistance of ChatGPT. It's evident from the graph that the presence of ChatGPT leads to a higher mean reward throughout the training steps, with the smoothed trends indicating a more consistent and possibly more efficient learning process. This analysis demonstrates that ChatGPT's contribution to the training framework significantly boosts the drone swarm system's efficiency in achieving its tasks.

Table. 1 The weights for rescue and search given by the ChatGPT

	Search 20% of the area.	Search 50% of the area	Search 80% of the area
Reach 0 target	[0.9, 0.1]	[0.8, 0.2]	[0.7, 0.3]
Reach 1 target	[0.6, 0.4]	[0.5, 0.5]	[0.4, 0.6]
Reach 2 targets	[0.3, 0.7]	[0.2, 0.8]	[0.1, 0.9]

Figure 4. Learning curves with GPT or not

CONCLUSIONS

In this research, we developed an innovative framework for drone swarms in urban search and rescue operations, utilizing dynamic weights in reinforcement learning. Our study addresses the crucial need for autonomy for the lack of effective methods to adjust priorities of conflicting goals in complex tasks such as post-disaster search and rescue. We integrated a large language model, such as ChatGPT, as a meta-agent within the reinforcement learning process. This model not only accelerates the training speed of the drone swarm agent model but also dynamically adjusts task priorities based on real-time environmental data, thereby enhancing decision-making. Our approach significantly boosts the efficiency and adaptability of drone swarms in multi-objective tasks within urban environments. This research represents a significant leap forward in the application of autonomous drone swarms for disaster response, providing a more responsive and agile solution for search and rescue operations.

As for future works, our primary aim is to critically assess and enhance the decision-making capabilities of ChatGPT within our reinforcement learning framework. While ChatGPT has served effectively as a meta-agent, managing a broad range of parameters and functions, its ability to make strategic decisions based on complex contextual information raises important concerns. Given its fundamental mechanism of predicting the next token based on previous ones, further research is needed to justify its suitability as a reinforcement learning agent. Furthermore, our simulations, currently focused on search area and target acquisition, do not adequately represent the unpredictable and complex nature of real-world environments. This limitation highlights the need for transitioning from controlled simulations to more realistic, dynamic settings that feature challenging obstacles and weather conditions. Such enhancements are crucial for preparing the system for real-world deployment, where unpredictability and a multitude of interacting variables are the norm. The next phase of our research will thus include extensive real-world testing. These tests are vital for refining our methodologies and rigorously validating the effectiveness of ChatGPT in actual search and rescue operations, ensuring our system's robustness and reliability in genuine emergency situations.

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